

Decentralization or Favoritism?

An Analysis of Ethereum Transactions and Maximal Extractable Value Strategies

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Abstract—Maximal Extractable Value (MEV) refers to the maximum value that can be extracted from block production beyond standard block rewards, by reordering, including, or excluding transactions in a block. This phenomenon is nowadays mainly connected to the increasing complexity of Decentralized Finance (DeFi) applications and the flexibility of blockchains like Ethereum. MEV can pose significant challenges to blockchain decentralization, particularly in Ethereum’s Proof-of-Stake ecosystem. This study aims to demonstrate these challenges by analyzing Ethereum, the most prominent blockchain where MEV extraction occurs. In fact, by analyzing the 220,993 blocks and therein 36,015,340 transactions from January 2024, this study reveals systematic favoritism-oriented behaviors in transaction ordering. MEV-related blocks demonstrate substantial transaction reordering, with significant evidence that the reordering often associates with favoring the transactions that were privately communicated to the block builders or even involved hidden payments made directly to those block builders. Moreover, we were most often able to track those transactions to speculative and malicious activities, characterized by very high economical values, and mostly encoded into bots that pursue MEV revenues in speculative fashion. Our findings suggest an emerging parallel economy that potentially undermines blockchain’s democratic principles, highlighting the need for governance mechanisms to preserve decentralization and user fairness.

Index Terms—Blockchain, Maximal Extractable Value (MEV), Ethereum, Decentralization, Blockchain democracy.

I. INTRODUCTION

The advent of blockchain technology has revolutionized the financial world. However, new challenges have emerged as these systems evolved, potentially threatening the decentralization and fairness that blockchains aim to uphold. One such challenge is *Maximal Extractable Value* (MEV), which has gained prominence in recent years. MEV refers to the maximum value that can be extracted from block production, beyond standard block rewards and gas fees, by including, excluding, or reordering transactions within a block. Although MEV can facilitate legitimate arbitrage and market efficiency, it also enables malicious practices that harm users and undermine the decentralization and democracy of blockchains. The shift of the Ethereum blockchain to *Proof-of-Stake* (PoS) has also led to the rise of an “off-chain” MEV ecosystem, involving different actors such as *searchers*, *builders*, and

relays. These participants manipulate transaction ordering for profit, exploiting collusion with block builders, thereby raising concerns about the impacts on blockchain democratization.

In this context, this paper analyzes MEV in Ethereum’s PoS system, focusing on transaction ordering patterns and speculative behaviors, which may indicate systemic biases or favoritism in block construction. Our research, in particular, is guided by the following two research questions:

- **RQ1:** *Does the ordering of transactions within blockchain blocks reveal any form of undemocratic favoritism?*
- **RQ2:** *Is there a link between the favoritism of the private transactions from the MEV ecosystem and speculative, possibly malicious behaviors?*

The remainder of the paper is organized as follows. Section II provides a background on Proof-of-Stake and MEV, explaining their interplay. Section III presents the MEV ecosystem in Ethereum, as a practical example of the momentum that is recently characterizing the MEV phenomenon, specifically for Proof-of-Stake blockchains. Section IV discusses the legitimate and illegitimate ways in which blockchain users can pursue MEV strategies, and outlines MEV attack typologies. Section V presents our methodology for data collection and our findings. Section VI reviews related work, situating our study in the broader context of blockchain and MEV research. Finally, Section VII concludes the paper and provides some perspectives on future research directions.

II. PROOF-OF-STAKE AND MAXIMAL EXTRACTABLE VALUE

PoS is a consensus mechanism that is used to decide which transactions, among those proposed by users, should be written on the blockchain blocks. In PoS blockchains, there are no miners in the traditional sense, as there are in *Proof-of-Work* (PoW); instead, *validators* are responsible for adding new blocks to the chain. Validators are chosen among those users willing to demonstrate ownership of a certain amount of cryptocurrency by putting it in a verifiable *stake*. In fact, in PoS, the probability of being chosen as a validator and therefore participating in the creation of the next block and receiving the associated reward is proportional to the amount

of cryptocurrency tokens they stake. If a validator is found trying to validate illegal transactions – for example, containing double spending – then the amount of tokens they put into the stake is slashed. On one side, this mechanism reinforces users’ trust in blockchain management. The underlying idea is that users with the highest stakes in the system have the greatest incentive to maintain the network’s security, as they would face the worst consequences if the cryptocurrency’s reputation and value declined due to attacks. On the other side, the collateralized cryptocurrency tokens staked by aspiring validators strengthen the economic model underpinning the blockchain system, providing it with greater stability and robustness.

However, the PoS consensus protocol has potential risks related to the proliferation of actors in the blockchain implementing *Maximal Extractable Value* (MEV) strategies [1]. MEV refers to the maximum amount of profit that a miner or block validator can extract from a given set of transactions in a blockchain, typically by reordering the transactions while building the blocks, or by including additional transactions in their blocks, such that they maximize the revenues that those blocks yield once they become part of the blockchain. MEV strategies can be pursued *on-chain*, by publicly proposing MEV-oriented transactions in the public *mempool* (the temporary storage for user-submitted transactions awaiting inclusion in the blockchain, visible to all network nodes), or *off-chain*, that is, based on actions that occur outside the blockchain, such as private negotiations of possible MEV transactions with miners or validators.

Off-chain solutions work particularly well with PoS blockchains, since the validator responsible for proposing the next block is designated one epoch in advance, allowing negotiations to take place at that time. In fact, after the introduction of PoS, Ethereum has experienced the rise of a rich off-chain MEV ecosystem, in which new actors also operate, such as *searchers*, *builders*, and *relays* (further detailed in Section III). Searchers identify MEV opportunities, builders further optimize MEV revenues by building as-profitable-as-possible blocks, and relays mediate the negotiation between builders and validators, by facilitating multiple builders to bid for the next block space and validators to select the most profitable blocks [2]. In fact, off-chain MEV strategies are less visible and harder to detect. They are not negative per se, and can be exploited for pursuing honest speculations, but can also incentivize malicious activities that can affect on-chain transactions (e.g., sandwich attacks), conceal non-democratic behaviors (e.g., censorship), and impact the network’s decentralization and security [3]. To mitigate the potential negative impact of private MEV strategies (especially off-chain) on Ethereum, a research and development organization known as *Flashbots* has started a set of initiatives to enable a permissionless, transparent, and fair ecosystem for MEV extraction [4].

III. EXAMPLE: THE MEV ECOSYSTEM IN ETHEREUM

As previously mentioned, the MEV ecosystem in Ethereum brings both economic opportunities and risks of centraliza-

tion to the network. To address these latter challenges, the *Proposer-Builder Separation* (PBS) mechanism has been introduced. PBS decouples block proposers (validators) from block builders, allowing specialized builders to optimize transaction ordering and MEV extraction, while validators focus on selecting the most profitable block. Specifically, PBS introduces the following distinct roles within the MEV ecosystem:

(i) *Searchers*. These actors identify and exploit MEV opportunities, such as arbitrage between *Decentralized Exchanges* (DEXs), sandwich attacks, and liquidations of under-collateralized loans (further detailed in Section IV). Searchers use sophisticated algorithms and bots to detect profitable scenarios and construct bundles of transactions with specific order requirements. To protect these strategies from being frontrun, searchers often use private transaction pools via relays such as Flashbots. This approach ensures privacy and minimizes the risk of strategies being copied. Searchers may also include users, like DEX traders, who seek frontrunning protection. Under PBS, searchers submit bundles of transactions that may combine private transactions with public ones, bidding competitively with higher gas fees or direct ETH payments to builders to secure inclusion in blocks. This bidding mechanism ensures that searchers maximize the value of their strategies and incentivizes builders to prioritize their bundles.

(ii) *Block builders*. These actors assemble blocks from various sources, including public mempool transactions, private bundles from searchers, and their own private transactions. The goal is to optimize each block’s profitability by carefully ordering transactions to extract MEV. Once a block is constructed, builders send it to relays, such as Flashbots, which hold the block in escrow until validators commit to it. Builders submit bids for inclusion, offering a portion of the MEV extracted to incentivize the validator to select their block. This competitive market fosters efficient block creation while decentralizing MEV extraction, ensuring that builders and validators benefit from these opportunities.

(iii) *Relays*. These actors connect searchers, builders, and validators within the PBS framework, serving as intermediaries that ensure fairness, privacy, and security. Flashbots is one such relay, which facilitates private bundle submission to mitigate harmful MEV effects such as frontrunning. Relays play a critical role in accepting blocks from builders and holding them in escrow. Initially, only the block header, which contains the most profitable block information, is sent to the validator. Once the validator commits to proposing the block, by signing the header, the full block is transmitted for inclusion in the blockchain. Relays maintain the confidentiality of the block’s contents until the validator commits, safeguarding the strategies employed by searchers and builders.

(iv) *Validators (Block proposers)*. Validators, responsible for proposing blocks to the network, have adapted to PBS by selecting the most profitable block provided by builders through relays. Validators may subscribe to multiple relays, such that those relays will send them the most profitable blocks constructed by the builders connected to each relay. The validators will then choose the highest-bid block based

on MEV rewards. The selected block must comply with Ethereum's consensus rules before being proposed to the network. Validators abstract the complexities of MEV extraction, focusing on selecting, signing, and proposing the block that maximizes their rewards. To participate in PBS, validators use the MEV-Boost client, allowing them to connect to relays and receive bids for block inclusion.

To sum up, the aim of PBS is to foster decentralization by creating competition among builders, mitigating harmful MEV effects through private transactions, and simplifying validators' roles. However, challenges remain, including risks of centralization among builders and relays, ecosystem complexity, and potential censorship concerns. Further innovation is needed to address these issues while supporting Ethereum's overarching goals of fairness and decentralization.

IV. MEV ATTACK TYPOLOGIES

The high profits that can be derived from speculatively adopting MEV strategies could incentivize *Decentralized Finance* (DeFi) traders to collude with block producers in order to exploit their inherent advantage in creating blocks that maximize MEV profits. Even worse, those collusions may aim to MEV strategies that maliciously penalize other blockchain users. The Flashbots organization refers to such phenomenon as the MEV *dystopia*, pointing out that it can seriously threaten the decentralization and democracy promised by permissionless blockchain platforms. Decentralization can be threatened because a few large DeFi firms may eventually take control, directly or indirectly, of the production of most blocks. Democracy can be threatened because the decisions made by block producers are driven by hidden logics of subjective interest (that derive from the collusions), rather than by transparent democratic rules.

In this paper, we address the problem of whether or not MEV profits are being currently pursued with collusive, hidden logics, and how this relates to participating in the off-chain MEV ecosystems and with pursuing malicious MEV strategies. To this end, we first outline a set of *principles* that should ideally characterize the non-collusive MEV strategies. We see these principles as necessary conditions for decentralization and democracy to be least affected by the MEV strategies:

- *Principle 1: No collusion.* Blockchain actors may pursue MEV strategies, provided that they operate independently from each other.
- *Principle 2: Meritocracy.* With respect to third parties who may or may not be pursuing MEV strategies, block producers act as economically rational actors, selecting the transactions (and the blocks) that pay the highest gas prices.
- *Principle 3: Legitimate MEV.* Blockchain actors shall pursue only legitimate, that is, non-malicious, MEV strategies.

The first two principles aim to guarantee decentralization and democracy, respectively. Pursuing a MEV strategy can be a legitimate behavior for blockchain actors, provided that it happens in a purely decentralized fashion, that is, without

collusion. It should be noted that the *no-collusion* principle also allows block builders (who are themselves blockchain actors) to pursue their own MEV strategies, such as by suitably reordering the transactions in the blocks they forge. However, it does not permit them to endorse the MEV strategies of third parties. Concerning third parties, they shall aim to guarantee maximal democracy, relying solely on purely rational decisions, as stated by the *meritocracy* principle. For example, it violates these two principles if a trader colludes with a block producer to obtain favoritism. In other words, it is reasonable for block producers to pursue their own MEV strategies, and it is also reasonable for them to profit indirectly, via transaction fees, from searchers and traders pursuing MEV strategies. Conversely, it is undemocratic for them to favor traders who provide hidden payments.

The third principle concerns distinguishing between types of MEV strategies that may be malicious or not. The *legitimate-MEV* principle forbids malicious MEV strategies, which generate profits for some blockchain users at the expense of others who are targeted as victims. In this paper, we specifically consider the following types of legitimate and malicious MEV strategies:

(i) *Decentralized EXchange (DEX) Arbitrage* is a very well-known, often very competitive MEV opportunity. If a token is offered at different prices in different DEX markets, one can buy the token on the lower-priced DEX and sell it on the higher-priced DEX. Moreover, by placing the two transactions (i.e., buy and sell) in the same block, or embedding them in a single smart contract call, blockchain traders can succeed in instantiating the ideal, riskless scenario in which both those transactions happen atomically. In legitimate scenarios, i.e., when decentralization and democracy are not affected based on the principles that we stated above, DEX arbitrages are *legitimate MEV strategies*.

(ii) *Liquidations* of collateral deposits from borrowers may be allowed under the occurrence of some conditions of market fluctuation. Upon asking to be liquidated, those borrowers pay a liquidation fee. Blockchain traders compete to be the first to submit a liquidation transaction in order to gain the liquidation fees paid by borrowers. In legitimate scenarios, liquidations are *legitimate MEV strategies*.

(iii) *Frontrunning* is a way to steal profitable transactions from other blockchain users. It is by-design intended as a *malicious strategy* for making a profit at the expense of victim users. A trader may become aware (e.g., by watching the transaction mempool) that some user is submitting a profitable transaction. If such a transaction depends only on who is first to submit it, then it can be stolen. This is, for example, the case of the DEX-arbitrage and Liquidation transactions we described above. The frontrunners will submit *copied transactions*, in which they substitute their own addresses in place of the victims' addresses, and will aim for those transactions to appear (in the forged blocks) before the original ones of the victims, thus stealing the corresponding profits.

(iv) *Sandwich* attacks take advantage of market fluctuations determined by some transactions of other blockchain

users. Traders may exploit the opportunities associated with becoming aware of those fluctuations in advance (e.g., by watching the transaction mempool) by inserting suitable sell-buy pairs of transactions from themselves, before and after the victims' transactions, respectively. It should be noted that, in most cases, the interference of the sandwich transactions may penalize the profitability of the victim transactions, and thus sandwich transactions are generally intended as a *malicious strategy*. In a sandwich transaction, the former transaction of the pair, which shall anticipate the targeted victim transaction, works in the same fashion as frontrunning. The latter transaction of the pair is called *backrunning*, by analogy.

It is common sense to expect that the legitimate-MEV principles outlined above are merely ideal and may sometimes be violated in practical scenarios. However, it is crucial to investigate the magnitude and effects of this phenomenon. In the analysis presented in the next section, we examined Ethereum blockchain blocks over a specific period, aiming to quantify these violations and assess how far we are from the ideal scenario.

V. A DEEP DIVE INTO ETHEREUM TRANSACTIONS

This section is dedicated to describing the data used and the analyses conducted to address the research questions underlying this work.¹

A. Data Collection

We collected all blocks of the Ethereum blockchain in January 2024, encompassing a total of 220,993 blocks (from block 18,908,895 to block 19,129,888) and 36,015,340 transactions therein. We used the *XBlock-ETH* datasets [5] to obtain data from the Ethereum blockchain, and the *Etherscan* [6] and *MetaMask* API [7] to analyze the block headers and the transactions in the block bodies. We then constructed a dataset that links each transaction to the following information:

(i) *Public* or *private*, that is, whether or not the transaction was collected via the public mempool. We used the tool *Mempool Dumpster*, provided by the Flashbots researchers [8], to gather the transaction hashes of those submitted to the public mempool. Transactions in our dataset that matched these hashes were classified as *public*. The remaining transactions were classified as *private*: if a transaction was not collected via the public mempool, we assume that the block producer obtained it through a private, off-chain channel, such as platforms operating within Ethereum's off-chain MEV ecosystem.

(ii) *MEV* or *NoMEV*, that is, whether the transaction was included in a block originating from a relay within the MEV ecosystem. Specifically, this was determined by checking if the block hash matched any block hashes we collected via the APIs of the following relays: *Aestus* [9], *Agnostic Gnosis* [10], *BloXroute MaxProfit* [11], *BloXroute Regulated* [12], *Eden Network* [13], *Flashbots* [14], *Manifold* [15], *Titan* [16], and

Ultrasound [17]. In our dataset, 90.6% of the blocks could be traced to these relays.

(iii) *Builder-payment*, that is, whether the transaction includes a payment to the builder of the block that contains it.

(iv) *Index*, that is, the position index of the transaction within its containing block, starting with one for the first transaction and incrementing sequentially for each subsequent transaction.

(v) *Gas price*, that is, the gas price paid by the transaction to the fee recipient of its block.

(vi) *Value*, that is, the economic value of the payment made in the transaction, in USD. We computed these values by gathering the USD prices of the tokens involved at the time of the transaction's inclusion in the blockchain, and then multiplying the number of tokens paid by the corresponding USD price.

(vii) *MEV role*, that is, the MEV-related role of the transaction, if any, classified as *arbitrage*, *liquidation*, *swap*, *frontrun*, *backrun*, *sandwich* or *none*. We used the Zeromev API [18] to collect the classification of the role of each transaction in our dataset. The labels *arbitrage*, *liquidation*, and *frontrun* directly correspond to the homonymous MEV strategies that we described in Section IV. The label *sandwich* refers specifically to transactions identified as victims of the sandwich attacks: as we described in Section IV, a sandwich attack consists of malicious *frontrun* and *backrun* transactions that suitably surround the victim transaction. The *swap* label identifies transactions that swap tokens with other tokens of different types, which is most often required when actuating MEV strategies.

The collection of payment data (related to the *builder-payment* and *value* data above) required additional processing for those transactions that represent smart contract calls, which we identified using the Etherscan API [6]. By inspecting the transaction receipts provided by Etherscan, we identified all activities triggered by the smart contracts that involved payments to blockchain accounts. If a paid account matched the block builder, we classified the transaction as *builder-payment*. Additionally, we assigned the *value* of the transaction as the maximum payment made by the smart contract, as we believe this serves as a suitable proxy for quantifying the monetary importance of the transactions involved.

We then partitioned the transactions in our dataset into disjoint groups based on combinations of their classifications as *public/private*, *MEV/NoMEV*, and *builder-payment*. However, as expected, four of the eight possible groups were empty or nearly empty. Two groups were empty because we did not find any *public* transactions that contained payments to builders. There are clear reasons why such collusive behavior is typically hidden in privately submitted transactions. Two other groups were nearly empty because we found only a negligible number (less than 0.01% of the dataset) of *private* transactions within *NoMEV* blocks. By design, validators who craft their own blocks (rather than selecting them from MEV ecosystem relays) typically choose transactions from the public mempool, with very few exceptions.

¹A replication package is available at <https://github.com/Davidejz/MEV-Data-Ethereum>

As a result, our dataset was divided into four disjoint groups, which we labeled as follows:

- *NoMEV-public* transactions. The mempool transactions included into blocks that did not originate from MEV relays: 2,872,835 transactions (8% of the dataset).
- *MEV-public* transactions. The mempool transactions included into blocks from MEV relays: 29,406,322 transactions (81.6% of the dataset).
- *MEV-private* transactions. Private transactions (from a MEV relay) that do not involve payment to the block builder: 3,460,670 transactions (9.6% of the dataset).
- *Builder-payment* transactions. Private transactions that involve payment to the builder: 275,513 transactions (0.8% of the dataset)

Transactions from the mempool represent the vast majority of the dataset, with most being incorporated into blocks crafted by the MEV ecosystem. Private transactions constitute a smaller portion, and an even smaller portion involves builder payments. However, we were particularly interested in the dynamics within the latter two groups to investigate how they differed from those of public transactions and whether these differences could reveal illegitimate behaviors.

B. Research Questions and Findings

We aim to address the research questions posed in the Introduction in this section. Specifically, the first question is:

- **RQ1:** *Does the ordering of transactions within blockchain blocks reveal any form of undemocratic favoritism?*

In Section IV, we argued that legitimate MEV should be pursued by blockchain participants without colluding with block producers, as such collusion could undermine the democracy and decentralization of the blockchain consensus. In principle, a sign of democracy is that block producers should process available transactions based solely on the gas prices those transactions pay. We then analyzed the extent to which the transactions in our dataset comply with this requirement. For each transaction, we quantified the shift of the actual ordering of the transaction in its block with respect to its ideal gas-price-based ordering as:

$$Tx_Shift = Actual_Tx_Index - Ideal_Tx_Index$$

where *Actual_Tx_Index* is the index of a transaction within its block, and *Ideal_Tx_Index* represents the hypothetical position that the transaction would occupy if the transactions in the block were sorted in descending order of their gas prices. The sign of the *Tx_Shift* value indicates whether a transaction was *prioritized* (if the sign is negative) or *delayed* within its block (if the sign is positive).

Figure 1 plots the distribution of the *Tx_Shift* values for the different groups of transactions in our dataset. Across the (all public) transactions that belong to NoMEV blocks (plot at the bottom of the figure), we observe an almost ideal distribution: excluding the outliers, the value of *Tx_Shift* is always zero, meaning that those transactions were picked in gas-price order. In contrast, for the transactions of the other three groups,

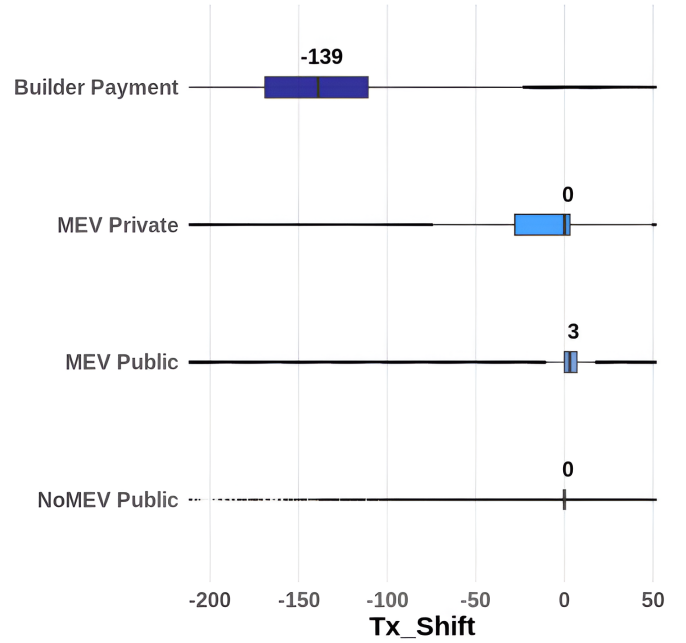


Fig. 1. Distributions of *Tx_Shift* for the *builder-payment*, *MEV-private*, *MEV-public*, *NoMEV-public* transactions in the dataset.

which all belong to MEV blocks, the distributions differ significantly.

In the *MEV-private* transactions (i.e., the transactions of MEV blocks that were privately collected by block producers, about 3.4 millions of transactions in our dataset), the distribution is skewed towards negative values, indicating that a relevant portion of them occurred with prioritized order. The first quartile of the *Tx_Shift* distribution is -28 .

In the *MEV-public* transactions (i.e., the transactions of MEV blocks that were collected from the public mempool, about 29.4 millions of transactions in our dataset) the median is positive (3), indicating that many transactions were delayed. Interestingly, the distributions of *MEV-private* and *MEV-public* transactions tend to opposite *Tx_Shift* values. This indicates that, within MEV blocks, the priority given to private transactions more often occurs at the expense of delaying public transactions.

Crucially, the transactions that involve payments to the block builder have strongly negative *Tx_Shift* values, with median centered at -139 . We interpret this as a clear sign of the existence of collusion (subject to payment) between the searchers that submitted those transactions and the builders that privileged the position of those transactions.

Figure 2 further details the differences in the orderings of the transactions that belong to the different groups, by displaying the distributions of the actual transaction index for each group. It is evident that both *builder-payment* and *MEV-private* transactions are consistently placed close to the beginning of the block, with respect to both the transactions in the other groups and the statistics of the entire dataset (the plot at the bottom of the figure, labeled as *population*). This

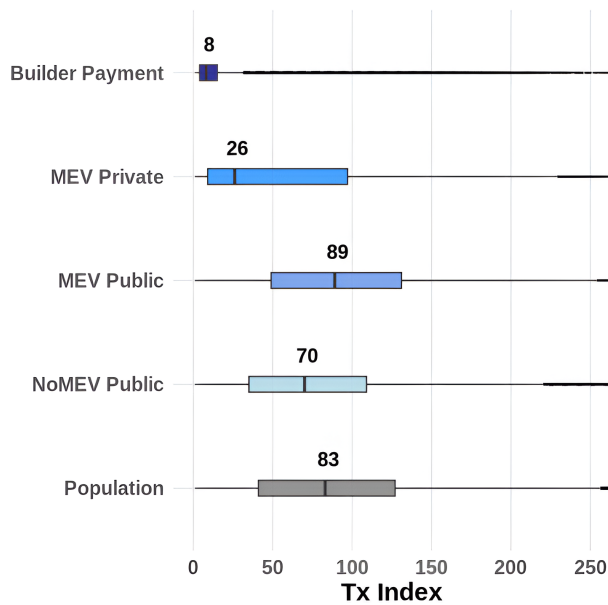


Fig. 2. Distributions of actual transaction index for *builder-payment*, *MEV-private*, *MEV-public*, *NoMEV-public* transactions in the dataset.

suggests a systematic prioritization of the private transactions. In contrast, transactions in the *NoMEV-public* category exhibit a distribution closely aligned with the ideal ordering, with the majority of transactions following a gas fee-based structure.

These findings indicate that the reordering of transactions is prevalent in MEV blocks, while NoMEV blocks follow a more predictable, gas-price-ordered structure without evidence of favoritism or manipulation. In particular, for both the *builder-payment* and *MEV-private* transactions, we observe a significant negative shift, indicating that these transactions not only frequently appear at the beginning of the block, but are also systematically favored in the ordering process, consistently being prioritized ahead of their neutral positions based on gas prices. This pattern highlights how these categories experience a form of preferential treatment, as their positions are adjusted to appear earlier than they would in a purely gas-price-ordered block. On the other hand, the *NoMEV-public* group shows no meaningful shift, reinforcing that transactions in these blocks maintain a more predictable and neutral order.

We now focus on the second research question:

- **RQ2:** *Is there a link between the favoritism of the private transactions from the MEV ecosystem and speculative, possibly malicious behaviors?*

To answer this question, we investigated to what extent the transactions in our dataset corresponded to profitable or malicious MEV activities, and the relation between those profitable or malicious MEV activities and the different groups of transactions considered in the previous research question.

Figure 3 visualizes the split of the transactions across the different types of MEV roles for both the entire dataset (first bar, labeled population) and the considered groups out of it (as explained earlier, we used the Zeromev API to

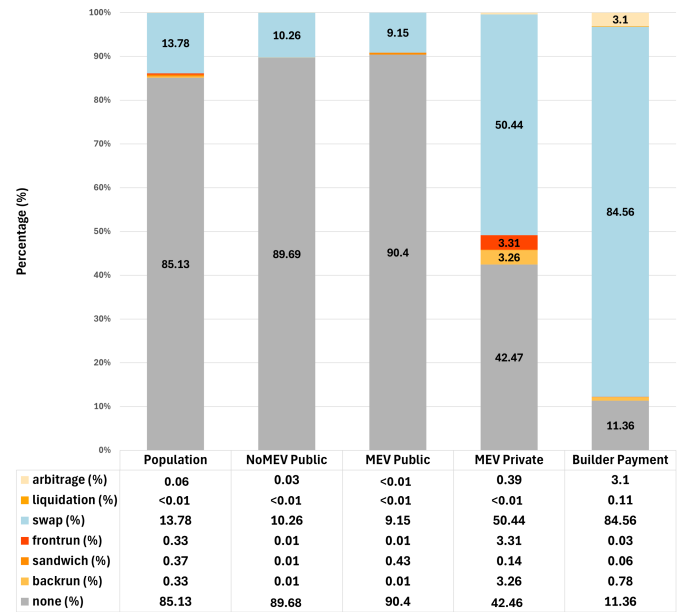


Fig. 3. MEV-role portions across the dataset, and the *builder-payment*, *MEV-private*, *MEV-public*, *NoMEV-public* transactions in the dataset.

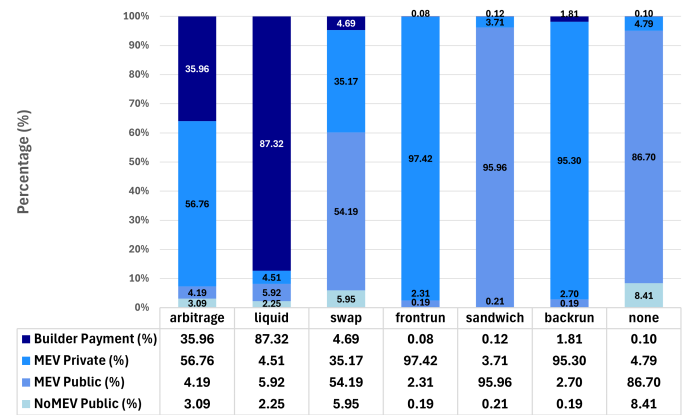


Fig. 4. Portions of *builder-payment*, *MEV-private*, *MEV-public*, *NoMEV-public* transactions for the transactions with each MEV role.

classify the MEV role of the transactions in our dataset). Figure 4 provides the orthogonal visualization which splits, for each type of MEV role, the portions that belong to different transaction groups.

Figure 3 reveals that malicious MEV activities, i.e., frontrun and backrun transactions (which comprise sandwich attacks), mostly occur in the private transactions of MEV blocks. Indeed, this can be expected, as malicious activities aim to hide themselves far away from the public mempool. Interestingly, a significant portion of arbitrages occur among the transactions of smart contracts that issue payments to the block builders. This can be due to the importance of placing those transactions timely, which likely stimulates traders to incentivize the collusive favoritism from block builders.

Figure 4 further highlights that significant portions of the highly remunerative MEV activities, such as arbitrage and

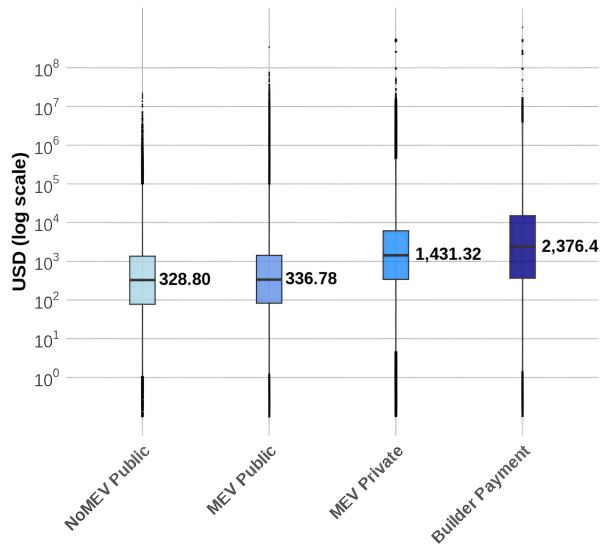


Fig. 5. Distribution of transaction values (in USD, log scale) across the considered transaction groups – considering the transactions with value greater than 0.1 USD.

liquidations, occur via smart contracts that involve payments to the builders. Furthermore, almost all frontrunning and backrunning activities are undertaken with private MEV transactions, whereas the victims of sandwich attacks are, in 96% of cases, public transactions.

These findings underscore the strong link between the MEV ecosystem and the malicious or collusive MEV activities. They reinforce the motivations of why these transaction types are systematically prioritized and influenced by external factors, deviating from a neutral gas-price-based ordering of the transactions within the blocks in the blockchain.

The analysis of the economic values of the transactions in each group further motivates the reasons for such a strong link. Figure 5 summarizes the distribution of the economic value (in USD) of the transactions in each group, as boxplots with y-axis in logarithmic scale. We filtered out the transactions with negligible values, that is, less than 0.1 USD. It is evident that the transactions in the *builder-payment* group involve the highest economic values, up to more than 10,000 USD at the third quartile of the distribution. In general, the transactions in the *MEV-private* group have significantly higher economic values than the ones in the groups of public transactions.

Further interesting considerations derive from analyzing the type of recipients of the transactions. We collected this information with the Etherscan API that classifies the recipients as smart contracts or EOAs (where EOA stands for externally owned account, i.e., an individual blockchain account). Other than generically labeled smart contracts, Etherscan further classifies two specific types of MEV-related smart contracts, i.e., MEV bots, which actively seek MEV opportunities, and DEX contracts, which allow interaction with DEX markets, as required for arbitrages and liquidations.

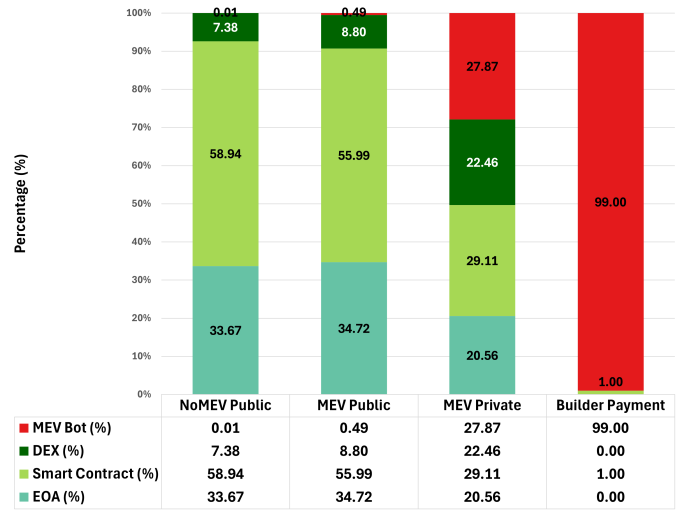


Fig. 6. Distribution of transaction recipient types across transaction groups.

Figure 6 visualizes the results. We observe that the vast majority of *builder-payment* transactions call MEV-Bot smart contracts, which also appear with a significant share among the recipients of *MEV-private* transactions in general. This makes it very evident that most colluding behaviors and many private transactions correlate with executing MEV strategies that were automated via suitable bots. We also observe that the smart contracts of DEX markets are called from both public and private transactions, but their share is significantly higher in the group of private transactions. This confirms that traders may often hide their speculative activities into private transactions, in order to protect their business. The public transactions, both the MEV and NoMEV ones, prevalently associate with generic accounts or generic smart contracts.

C. Limitations

In our analysis we relied on the assumption that, ideally, block producers should process available transactions based solely on the corresponding gas fees. However, this assumption might not necessarily hold in all cases. For instance, other authors propose to compute the ideal index of a transaction based on the greedy sequencing rule [19]. In future work, we plan to extend our results with respect to other possible ideal orderings of transactions.

Our current analysis is confined to Ethereum's blocks of January 2024, which limits the generalizability of our findings for longer time horizons or different blockchain ecosystems.

VI. RELATED WORK

Daian et al. [20] first highlighted Miner Extractable Value (MEV) in decentralized exchanges, where miners could re-order transactions to frontrun others, destabilizing the ecosystem. This set the stage for research on MEV's impact on blockchains. In 2022, Piet et al. [21] expanded on this, emphasizing how miners exploited transaction ordering to extract value from DeFi applications. Later that year, Qin et al. [22] quantified MEV's destabilizing effects and introduced

models to measure its economic consequences, urging further investigation into mitigating solutions.

As Ethereum transitioned to Proof-of-Stake (PoS) in 2023, several researchers engaged in investigating the impacts of MEV in Ethereum and the relations with the transition to PoS. Wahrstätter et al. [23] analyzed its potential to create new MEV extraction avenues, particularly through staking and block construction. Heimbach et al. [24] examined Proposer-Builder Separation (PBS), noting challenges in implementation. Öz et al. [25] highlighted the role of timing in MEV extraction, complicating mitigation efforts in PoS. Mancino et al. [26] explored off-chain MEV strategies, warning of centralization risks. Öz et al. [27] compared Ethereum's PoS with Algorand's consensus mechanism and identified key differences in how MEV could manifest in both systems, with implications for transaction fairness and decentralization in blockchain networks.

The work reported in this paper provides novel empirical data on the impacts of MEV in Ethereum. It is unprecedented in revealing that block builders may be arbitrarily favoring transaction orderings that are beneficial for the private transactions of the off-chain MEV ecosystem, which we see as a crucial sign of non-decentralized, non-democratic governance of the blockchains. In our dataset those favoritism behaviors also appear to correlate with the profitability of the (possibly malicious) MEV strategies, and with the implementation of those strategies in the form of MEV bots that include hidden payments for the block builders.

Some research efforts have recently addressed possible countermeasures for contrasting the negative effects of the MEV phenomenon. Yang et al. discussed 32 MEV mitigation strategies categorized into four main classes, i.e., MEV auction platforms, time-based ordering properties, content-agnostic ordering, and MEV-aware application design [28]. Their taxonomy highlights how different strategies balance security, efficiency, and decentralization concerns. MEV auction platforms, such as MEV-Boost [2], aim to make MEV extraction as transparent as possible, but may also reinforce centralization. Time-based ordering properties, such as Aequitas [29] and Themis [30], require that transactions get processed according to their arrival times, to prevent manipulation of the ordering. Content-agnostic ordering exploits encryption techniques (e.g., like in Shutter Network [31]) to obscure the transaction details at ordering time. MEV-aware application designs, such as CoWSwap [32] and FairTraDEX [33], integrate protocol-level mechanisms to contrast frontrunning and sandwich attacks.

Furthermore, the issues that arise from PBS and transaction reordering have been addressed in additional research efforts, which also proposed solutions to reduce MEV's broader negative externalities. Piet et al. introduced MEVade, a system to reduce MEV by optimizing block construction and transaction ordering [34]. Kavousi et al. proposed the Blind-Perm approach, which encrypts the mempool and permutes transaction orders to prevent MEV extraction [35]. Vedula et al. introduced Masquerade, a lightweight system to prevent transaction reordering [36]. Nasrulin et al. proposed a protocol

ensuring accountability in transaction sources [37]. Churiwala et al. presented the CoMMA Protocol, offering a secure solution for transaction ordering [38]. Bahrani et al. theoretically analyzed the centralization risks associated with block building and proposer-builder separation, by using game-theory and stochastic models [39].

Despite the above studies, a complete solution for MEV remains elusive. Our study properly complements this recent research body by providing empirical evidence of how MEV extraction impacts the transaction ordering in Ethereum.

VII. CONCLUSION

This study provides comprehensive evidence of systematic favoritism and speculative behaviors within the Ethereum blockchain's MEV ecosystem. Our analysis of transaction data from January 2024, covering over 220,000 blocks and more than 36 million transactions, reveals several significant findings that challenge the core principles of decentralization and transaction ordering in blockchain systems. Our analysis reveals three key findings. First, MEV blocks (90.6% of the total) exhibit significant deviation from gas price-based ordering, while NoMEV blocks maintain strict adherence to it. Second, we provide empirical evidence that *builder-payment* and *MEV-private* transactions are systematically prioritized by block builders, suggesting the existence of hidden collusion between traders and block builders. Third, these prioritized transactions are strongly correlated with speculative MEV activities such as arbitrage and liquidations, as well as malicious activities like frontrunning and sandwich attacks, which are often characterized by very high economic values and interactions with MEV bots.

Our findings indicate concerning implications. They reveal a parallel economy operating through hidden mechanisms, designed to systematically prioritize certain transactions. The prevalence of speculative behaviors suggests that the MEV ecosystem may be incentivizing practices detrimental to regular users. Future research should focus on developing mechanisms to better align MEV-related activities with decentralization principles, including new consensus mechanisms and user protection systems. Our research underscores how, without proper governance, MEV-related practices could undermine the fundamental decentralized nature of blockchain platforms.

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REFERENCES

- [1] Ethereum.org, “Maximal Extractable Value (MEV),” <https://ethereum.org/en/developers/docs/mev/>, 2023, accessed on December, 5th, 2023.

- [2] Flashbots, “Flashbots MEV-Boost - Introduction,” <https://docs.flashbots.net/flashbots-mev-boost/introduction>, 2022, accessed on December, 5th, 2024.
- [3] Flashbots, “MEV-Boost Risks and Considerations,” <https://docs.flashbots.net/flashbots-mev-boost/architecture-overview/risks>, 2022, accessed on December, 5th, 2024.
- [4] Flashbots, “Flashbots,” <https://www.flashbots.net/>, 2022, accessed on December, 5th, 2024.
- [5] P. Zheng, Z. Zheng, J. Wu, and H.-N. Dai, “Xblock-eth: Extracting and exploring blockchain data from Ethereum,” *IEEE Open Journal of the Computer Society*, vol. 1, pp. 95–106, 2020.
- [6] Etherscan, “Etherscan APIs documentation,” <https://docs.etherscan.io/>, 2024, accessed on December, 5th, 2024.
- [7] MetaMask, “MetaMask developer documentation,” <https://docs.metamask.io/services/reference/ethereum/json-rpc-methods/>, 2024, accessed on December, 5th, 2024.
- [8] Flashbots, “Mempool Dumpster,” <https://mempool-dumpster.flashbots.net/index.html>, 2024, accessed on December, 5th, 2024.
- [9] Aestus, “Aestus Relay,” <https://mainnet.aestus.live/>, 2022, accessed on December, 4th, 2024.
- [10] Agnostic Relay, “Agnostic Relay,” <https://agnostic-relay.net/>, 2022, accessed on December, 4th, 2024.
- [11] bloXroute Max Profit, “bloXroute Max Profit Relay,” <https://bloxroute.max-profit.blxrbdn.com/>, 2022, accessed on December, 4th, 2024.
- [12] bloXroute Regulated, “bloXroute Regulated Relay,” <https://bloxroute.regulated.blxrbdn.com/>, 2022, accessed on December, 4th, 2024.
- [13] Eden Network, “Eden Relay,” <https://docs.edennetwork.io/eden-relay/overview/>, 2024, accessed on December, 5th, 2024.
- [14] Flashbots, “Flashbots Relay,” <https://boost-relay.flashbots.net/>, 2022, accessed on December, 4th, 2024.
- [15] Manifold, “SecureRpc Relay,” <https://mainnet-relay.securerpc.com/>, 2022, accessed on December, 4th, 2024.
- [16] Titan, “Titan Global Relay,” <https://global.titanrelay.xyz/>, 2024, accessed on December, 4th, 2024.
- [17] Ultrasound, “Ultrasound Relay,” <https://relay.ultrasound.money/>, 2022, accessed on December, 4th, 2024.
- [18] Zeromev, “Zeromev,” <https://zeromev.org/>, 2024, accessed on December, 5th, 2024.
- [19] M. V. Xavier Ferreira and D. C. Parkes, “Credible decentralized exchange design via verifiable sequencing rules,” in *Proceedings of the 55th Annual ACM Symposium on Theory of Computing*, ser. STOC 2023. New York, NY, USA: Association for Computing Machinery, 2023, p. 723–736. [Online]. Available: <https://doi.org/10.1145/3564246.3585233>
- [20] P. Daian, S. Goldfeder, T. Kell, Y. Li, X. Zhao, I. Bentov, L. Breidenbach, and A. Juels, “Flash boys 2.0: Frontrunning in decentralized exchanges, miner extractable value, and consensus instability,” in *2020 IEEE Symposium on Security and Privacy (SP)*. IEEE Computer Society, 2020, pp. 910–927.
- [21] J. Piet, J. Fairroze, and N. Weaver, “Extracting godl [sic] from the salt mines: Ethereum miners extracting value,” 2022. [Online]. Available: <https://arxiv.org/abs/2203.15930>
- [22] K. Qin, L. Zhou, and A. Gervais, “Quantifying blockchain extractable value: How dark is the forest?” in *2022 IEEE Symposium on Security and Privacy (SP)*. IEEE, 2022, pp. 198–214.
- [23] A. Wahrstätter, L. Zhou, K. Qin, D. Svetinovic, and A. Gervais, “Time to bribe: Measuring block construction market,” *Cryptology ePrint Archive*, Paper 2023/760, 2023. [Online]. Available: <https://eprint.iacr.org/2023/760>
- [24] L. Heimbach, L. Kiffer, C. Ferreira Torres, and R. Wattenhofer, “Ethereum’s proposer-builder separation: Promises and realities,” in *Proceedings of the 2023 ACM on Internet Measurement Conference*, ser. IMC ’23. New York, NY, USA: Association for Computing Machinery, 2023, p. 406–420. [Online]. Available: <https://doi.org/10.1145/3618257.3624824>
- [25] B. Öz, B. Kraner, N. Vallarano, B. S. Kruger, F. Matthes, and C. J. Tessone, “Time moves faster when there is nothing you anticipate: The role of time in MEV rewards,” in *Proceedings of the 2023 Workshop on Decentralized Finance and Security*, ser. DeFi ’23. New York, NY, USA: Association for Computing Machinery, 2023, p. 1–8. [Online]. Available: <https://doi.org/10.1145/3605768.3623563>
- [26] D. Mancino, A. Leporati, M. Viviani, and G. Denaro, “A role and reward analysis in off-chain mechanisms for executing MEV strategies in Ethereum proof-of-stake,” *Distrib. Ledger Technol.*, Jun. 2024. [Online]. Available: <https://doi.org/10.1145/3672405>
- [27] B. Öz, J. Gebele, P. Singh, F. Rezabek, and F. Matthes, “Playing the MEV game on a first-come-first-served blockchain,” in *2024 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, 2024, pp. 220–224.
- [28] S. Yang, F. Zhang, K. Huang, X. Chen, Y. Yang, and F. Zhu, “Sok: MEV countermeasures,” in *Proceedings of the Workshop on Decentralized Finance and Security*, ser. DeFi ’24. New York, NY, USA: Association for Computing Machinery, 2024, p. 21–30. [Online]. Available: <https://doi.org/10.1145/3689931.3694911>
- [29] M. Kelkar, F. Zhang, S. Goldfeder, and A. Juels, “Order-fairness for byzantine consensus,” in *Advances in Cryptology – CRYPTO 2020: 40th Annual International Cryptology Conference, Santa Barbara, CA, USA, August 17–21, 2020, Proceedings, Part III*. Berlin, Heidelberg: Springer-Verlag, 2020, p. 451–480. [Online]. Available: https://doi.org/10.1007/978-3-030-56877-1_16
- [30] M. Kelkar, S. Deb, S. Long, A. Juels, and S. Kannan, “Themis: Fast, strong order-fairness in byzantine consensus,” in *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*, ser. CCS ’23. New York, NY, USA: Association for Computing Machinery, 2023, p. 475–489. [Online]. Available: <https://doi.org/10.1145/3576915.3616658>
- [31] Shutter Network, “Shutter network blog,” <https://blog.shutter.network/>, 2025, accessed on March, 22nd, 2025.
- [32] CoW Protocol, “Cow swap,” <https://swap.cow.fi/>, 2025, accessed on March, 22nd, 2025.
- [33] C. McMenamin, V. Daza, M. Fitzi, and P. O’Donoghue, “Fairtrader: A decentralised exchange preventing value extraction,” in *Proceedings of the 2022 ACM CCS Workshop on Decentralized Finance and Security*, ser. DeFi’22. New York, NY, USA: Association for Computing Machinery, 2022, p. 39–46. [Online]. Available: <https://doi.org/10.1145/3560832.3563439>
- [34] J. Piet, V. Nair, and S. Subramanian, “Mevade: An MEV-resistant blockchain design,” in *2023 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, 2023, pp. 1–9.
- [35] A. Kavousi, D. V. Le, P. Jovanovic, and G. Danezis, “BlindPerm: Efficient MEV mitigation with an encrypted mempool and permutation,” *Cryptology ePrint Archive*, Paper 2023/1061, 2023. [Online]. Available: <https://eprint.iacr.org/2023/1061>
- [36] A. Vedula, S. B. Venkatakrishnan, and A. Gupta, “Masquerade: Simple and lightweight transaction reordering mitigation in blockchains,” 2023. [Online]. Available: <https://arxiv.org/abs/2308.15347>
- [37] B. Nasrulin, G. Ishmaev, J. Decouchant, and J. Pouwelse, “LO: an accountable mempool for MEV resistance,” in *Proceedings of the 24th International Middleware Conference, Middleware 2023, Bologna, Italy, December 11–15, 2023*. ACM, 2023, pp. 98–110. [Online]. Available: <https://doi.org/10.1145/3590140.3629108>
- [38] D. Churiwala and B. Krishnamachari, “Comma protocol: Towards complete mitigation of maximal extractable value (MEV) attacks,” in *2023 Fifth International Conference on Blockchain Computing and Applications (BCCA)*, 2023, pp. 359–360.
- [39] M. Bahrani, P. Garimidi, and T. Roughgarden, “Centralization in block-building and proposer-builder separation,” in *Financial Cryptography and Data Security*, J. Clark and E. Shi, Eds. Cham: Springer Nature Switzerland, 2025, pp. 331–349.